# Magnet Man - Technical Design Document

Project Overview: This project is a first-person 3D puzzle-platformer built around a simplified simulation of magnetic forces. The player manipulates polarity-based interactions to navigate structured puzzle environments. The goal is to implement a real-time force-based movement system inspired by electromagnetism within a playable level-based game.

## Technology Stack Engine:

- Unity
- Language: C#
- Version Control: Git + GitHub
- Target Platform: Windows PC (primary)

### Core Systems Architecture

Modular system design including Player Controller, Magnetic Force System, Interaction System, Puzzle System, Level Management, UI, and VFX/Audio systems.

### Player Controller System

First-person movement with mouse look, jumping, and external force integration. Must remain responsive under physics-based magnetic forces.

### Magnetic Force System

Core gameplay system simulating simplified electromagnetic attraction and repulsion. Uses inverse-square inspired model $F \propto (q1*q2)/r^2$ applied in Unity FixedUpdate with force clamping for stability.

### Interaction System

Raycast-based detection of magnetic objects, polarity toggling, and interaction mode switching between self and object targeting. Puzzle System Linear room-based progression. Each room introduces or combines mechanics such as pressure plates, switches, and magnetic traversal challenges.

### Level Management

Scene-based or single-scene architecture using Unity SceneManager for progression and checkpoints. UI System Minimal HUD including polarity indicator, crosshair, and optional interaction prompts. VFX and Audio Particle systems and audio feedback for polarity changes, magnetic field activation, and movement impacts.

### Engineering Constraints

No multiplayer, procedural generation, full Maxwell simulation, or complex AI systems. Focus will be on stable, deterministic gameplay systems.

**Performance Goals**
Target of 60 FPS on mid-range hardware. Physics optimized via FixedUpdate and capped force calculations.

**Expected Deliverable**
A fully playable 3D puzzle level (5–8 rooms), functional electromagnetic system, and stable first-person controller with basic UI and feedback systems.

## Rooms

**Room 1: Repulsion**
Players learn about like-charge repulsion by propelling themself across a gap.

**Room 2: Attraction**
Players learn about opposite charge attraction by changing their polarity, drawing themself towards objects.

**Room 3: Switching Charges**
Reinforcing the ideas of room 1 and 2 via a more complex puzzle / maze

**Room 4: Force vs. Distance**
Players learn about how electric forces weaken or strengthen based on distance

**Room 5: Electric Field Chamber**
Learn about how charged particles move through electric fields, as the whole room contains an electric field that players must navigate.

**Room 6: Multiple Fields**
A more complex version of the previous room, containing multiple distinct electric fields that players must navigate

**Room 7: Magnetic Field Introduction**
An introduction to how velocity and magnetic fields change the direction of a charged particle

**Room 8: Final Test Chamber**
A combination of all previous concepts